

Nonuniform Λ -Distributions on Symmetric Groups

Cycle transforms, shuffles, and Fourier operators

Proof and computational verification note

17 July 2026

Outcome. Explicit permutation matrices for S_3, S_4, S_5 were compared with the cycle-count formula under six families of nonuniform measures. A separate S_3 calculation decomposes every tested tensor-symmetric character into all three irreducibles and verifies both central and noncentral Fourier operators. All **267** checks pass; the maximum absolute error is 4.45×10^{-15} .

1 Permutation matrices and the cycle transform

For the defining permutation representation $V = \mathbb{C}^n$, a d -cycle contributes all d th roots of unity. Consequently

$$\det(I - tP_\pi) = \prod_{d \geq 1} (1 - t^d)^{C_d(\pi)}. \quad (1)$$

With

$$Q_d(\mathbf{t}) = \prod_a (1 - t_a^d)^{-1}, \quad (2)$$

direct substitution in the determinant definition proves

$$\mathcal{S}_{\nu, n}(\mathbf{t}) = \mathbb{E}_\nu \left[\prod_{d \geq 1} Q_d(\mathbf{t})^{C_d(\pi)} \right]. \quad (3)$$

This identity is valid for every probability measure, whether or not it is a class measure. It is the joint cycle-count probability-generating function evaluated at the transformed variables Q_d .

For coefficient extraction, each factor is expanded using

$$(1 - t^d)^{-c} = \sum_{j \geq 0} \binom{c + j - 1}{j} t^{dj}. \quad (4)$$

The verification uses integer dynamic programming for (4), separately for every permutation. This route does not diagonalize a matrix.

2 Measures tested

All probabilities are constructed on the fully enumerated group:

1. the lazy random-transposition walk with $\kappa = 0.35\delta_e + 0.65 \text{Unif}(C_2)$, after three steps;
2. the lazy random 3-cycle walk with laziness 0.4, after two steps;
3. the lazy adjacent-transposition walk with laziness 0.25, after four steps;

4. the inverse 2-riffle law

$$\mathbb{P}(\pi) = 2^{-n} \binom{n+1-d(\pi)}{n}, \quad (5)$$

where $d(\pi)$ is the number of descents;

5. Ewens measure with $\theta = 1.7$, $\mathbb{P}(\pi) \propto \theta^{K(\pi)}$;

6. the general cycle weight $\mathbb{P}(\pi) \propto 1.2^{C_1} 0.7^{C_2} 1.6^{C_3}$.

For S_3, S_4, S_5 and every partition through total degree four, the direct route averages characters computed from the explicit matrices using Newton's identities. The cycle route applies (4) to the cycle structure. Thus the two routes share the probability law but not the spectral computation.

The normalizing generating-function formulas follow immediately from the exponential formula for labeled cycles. For example,

$$\mathcal{S}_{n,\theta}^{\text{Ewens}} = \frac{[z^n] \exp\left(\theta \sum_{d \geq 1} Q_d z^d / d\right)}{[z^n] \exp\left(\theta \sum_{d \geq 1} z^d / d\right)}. \quad (6)$$

Equation (6) records the corrected typography: the normalization is $(\theta)_n$, and the summation index is $d \geq 1$.

3 Fourier proof and an independent S_3 test

Decompose

$$W_\tau = \bigotimes_a \text{Sym}^{\tau_a} V \cong \bigoplus_{\lambda \vdash n} M_{\tau,\lambda} \otimes V_\lambda, \quad m_{\tau,\lambda} = \dim M_{\tau,\lambda}. \quad (7)$$

Linearity of trace gives, for an arbitrary measure,

$$[\mathbf{t}^\tau] \mathcal{S}_{\nu,n} = \sum_{\lambda \vdash n} m_{\tau,\lambda} \text{Tr } \hat{\nu}(\lambda), \quad \hat{\nu}(\lambda) = \sum_{\pi} \nu(\pi) \rho_\lambda(\pi). \quad (8)$$

For a convolution walk, $\widehat{\kappa^{*r}}(\lambda) = \widehat{\kappa}(\lambda)^r$. If κ is central, Schur's lemma makes this operator scalar. In the lazy transposition case,

$$\widehat{\kappa}(\lambda) = \beta_\lambda I, \quad \beta_\lambda = a + (1-a) \frac{\chi^\lambda((12))}{d_\lambda}. \quad (9)$$

For adjacent transpositions the full matrix power must be retained.

The S_3 test constructs the trivial, sign, and two-dimensional standard matrices explicitly. Multiplicities in (7) are obtained independently by the uniform character inner product, then (8) is evaluated for an arbitrary law, lazy central and adjacent walks, and the riffle law. For laziness $a = 0.35$, the central scalars on $(\mathbf{1}, \text{sgn}, \text{std})$ are

$$(1, -0.3, 0.35). \quad (10)$$

The adjacent operator on the standard representation instead has two distinct eigenvalues 0.025 and 0.675, directly exposing the claimed matrix-valued behavior.

4 Results

There are 12 partitions through total degree four. The cycle calculation therefore contributes $3 \cdot 6 \cdot 12 = 216$ comparisons. The four S_3 Fourier laws contribute 48 more, and the three irreducible operator checks bring the total to 267.

Case	coefficient	matrix/direct	proposed formula	result
S_3 , inverse 2-riffle	(3)	5.750000	5.750000	PASS
S_4 , Ewens $\theta = 1.7$	(2, 1)	7.429557	7.429557	PASS
S_5 , transposition walk	(2, 2)	39.683180	39.683180	PASS
S_3 , arbitrary Fourier law	(2, 1)	1.904762	1.904762	PASS
S_3 , adjacent walk Fourier law	(2, 2)	9.671793	9.671793	PASS

5 Warnings, corrections, and scope

Without laziness, a transposition walk alternates parity and does not converge to uniform measure on S_n . A walk on odd-length cycles stays in A_n . These are support obstructions, not failures of (3) or (8).

The supplied continuous-time class-walk expressions contained a stray comma in the exponential; the intended factor is $\exp(t(r_\lambda - 1))$. Continuous-time graph interchange processes were not numerically tested here, but their operator formula follows from replacing matrix powers in (8) by the matrix exponential of the generator.

Run

```
.venv/bin/python -m \
  for_this_guy.nonuniform_lambda_distributions.symmetric_groups.verification
```

or open `symmetric_groups/notebook.py` in marimo.